COMPX520 / 2026

Interim Report

Feel the Sports: Acoustic-Based Event Detection with Haptic Feedback for Racket Sports Spectators

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1 Introduction

1.1 Problem Definition

Watching a live sports match in person is a rich sensory experience. Spectators can clearly feel the physical energy of the game, especially during hard hits and fast, intense rallies. When people watch sports remotely through a screen, most physical experience is missed. They can only see and hear the game; they cannot 'feel' it.

This project is part of a broader "Feel the Sports" theme, which explores how technology can bring physical sensations back to remote sports spectators for enhancing the experience. The main challenge is detecting meaningful events and patterns from a sports video, such as the moment of a ball impact, a bounce, or the overall rhythm and pace of play, and using those detected signals to drive physical feedback devices worn or held by the viewer.

Traditional approaches rely on physical sensors such as accelerometers and IMUs placed on the player or equipment, which require hardware setup on the court and cannot be applied to existing recordings (Yin et al., 2021). Cameras and microphones offer richer, non-intrusive alternatives: a camera captures visual information while a microphone captures the acoustic events of the game. This project focuses on audio as the primary sensing modality, as it can extract ball events and rhythmic patterns from any existing sports video without hardware on the field of play (Baughman et al., 2019; Caprioli et al., 2024, 2025).

1.2 Aims of the Project

The aim of this project is to analyse the audio track of a sports video to detect ball events — specifically ball impact with the racket, and use these detections to drive vibrotactile feedback via an Android smartphone held by the spectator. The detections act as "virtual sensors" that replace physical hardware on the court or woren by the players.

If this aim is completed well ahead of schedule, a proof-of-concept extension could be explored: deriving game rhythm from detected events and driving a slow thermal signal via the Embr Wave 2 wristband. This would be a working prototype only — no user study would be conducted for it. Given the slow nature of thermal actuation and current API limitations, this extension is considered unlikely within the project timeline.

The actuator for this project is an **Android smartphone**, which vibrates in response to each detected event. Together, the detection system and the smartphone form an end-to-end "Feel the Sports" pipeline: *sports video audio → event detection → vibration feedback → spectator feelings.*

The specific sport to focus on is not assumed in advance. A formal comparison of racket sports is presented in Section 3 to justify the final selection based on temporal and engineering criteria.

2 Background

2.1 Haptic Feedback Systems for Sports Spectators

A common sense in haptic sports feedback systems is the use of event-driven triggering: a detected signal drives a feedback actuator. Systems differ mainly in how the signal is detected, what signal is used and what actuator delivers the feedback. Iekura et al. (2015) detected tool sounds and player heartbeat using microphones; Handa et al. (2019) tracked ball position with a camera system; and Peng (2022) captured biosensor signals (heart rate, muscle tension, skin temperature) from an experienced fan and replayed them to non-fans. These three approaches cover the main sensing strategies: acoustic, visual, and physiological.

On the actuator side, most systems use dedicated hardware that is not available to ordinary spectators. Custom vibration devices (Mizushina et al., 2015; Takahashi et al., 2021), Peltier thermal modules (Peng, 2022), and multi-point body vibration suits (de Graaf, 2022) deliver strong perceptual effects but require physical setup. Even systems targeting athletes rather than spectators — such as IMU-driven racket feedback (Sabnis et al., 2025) or vibration-based breathing guidance in VR exergaming (Greinacher et al., 2020) — confirm that vibration is an effective feedback channel in physical sports contexts.

Many studies show that sports viewers really like haptic feedback. However, it is hard for most people to use these systems because they usually require special, custom hardware. To address this, recent studies have started using consumer-grade devices. For example, Gómez-Monroy et al. (2026) sent event-triggered feedback using a standard smartphone without any extra equipment. his approach directly aligns with the present project. Since 70.1% of people around the world now own a smartphone (SQ Magazine, 2026), the spectator already holds the necessary hardware while watching the event.

2.2 The Embr Wave 2 as a Proof-of-Concept Extension

The Embr Wave 2 is a wrist-worn thermal wearable that delivers warming or cooling sensations via thermoelectric modules (Embr Labs, 2021). It is noted here as a candidate for a proof-of-concept extension only: thermal actuation is inherently slow (several seconds per cycle), — heating and cooling cycles take several seconds to be perceived — which makes tight synchronisation with fast sports events very difficult. These constraints make it unsuitable as a core deliverable.

2.3 Acoustic Event Detection in Tennis

Audio analysis is a reliable basis for detecting ball events in tennis because the acoustic signatures of impacts and bounces are relatively stable across matches, whereas video varies greatly due to camera angles and lighting (Baughman et al., 2019). Unlike physical sensor approaches, audio can be extracted from any existing broadcast recording without hardware on the court.

The processing approaches used in this area form a progression from simple to complex. Early work applied peak energy detection directly to the filtered waveform (Caprioli et al., 2024), which is sufficient when impacts are louder than background noise. More recent work added machine learning classification to handle noisier conditions: Baughman et al. (2019) used MFCCs with delta features as input to a CNN, achieving an F1-score of 92.39%; Caprioli et al. (2025) compared Decision Tree, Random Forest, SVM, and XGBoost, finding that XGBoost alone was sufficient for impact detection (median cross-validation accuracy 0.98), while the harder bounce detection required a soft-voting ensemble of XGBoost, SVM, and MLP (median accuracy 0.97).

A consistent challenge across all studies is noise. All three works tested their systems with ambient noise present — voices, wind, and adjacent court sounds — and found that bounce detection is more affected than impact detection, because the bounce sound is lower in amplitude (Caprioli et al., 2024, 2025). In real-court conditions, overall system accuracy dropped to approximately 85% (Caprioli et al., 2025). Audio from broadcast recordings adds further noise in the form of crowd and commentary, which none of these studies specifically addressed — this is a gap that the present project will encounter and report on.

2.4 Sound Features for Event Classification

Studies have shown that features such as MFCCs, zero-crossing rate, and spectral statistics extracted from short audio windows can reliably distinguish ball impacts, bounces, and background noise in racket sports recordings (Baughman et al., 2019; Caprioli et al., 2025).

2.5 Gaps Addressed by This Project

The existing studies focus on sports performance and analysis rather than spectator experience. Existing haptic spectator systems rely on custom hardware or dedicated physical devices. No existing work connects audio-based tennis event detection with accessible haptic feedback using consumer devices. This project addresses these gaps.

3 Sport Selection

Choosing the right sport is not only a practical decision but also an engineering one. In haptic system design, the temporal spacing between feedback events directly affects whether users can perceive and distinguish each impulse. This section provides a cross-sport comparison to justify the final choice.

3.1 The Inter-Shot Interval as a Haptic Design Constraint

In sports biomechanics and notation analysis, the time between one player striking the ball or shuttlecock to the opponent striking it back is designated as the **Inter-Shot Interval** (ISI) (Fernandez-Fernandez et al., 2009; Laffaye et al., 2015). Some papers call it **Rally Pace**, which reflects the same concept. It can be derived from match data as:

$$\text{ISI} = \frac{\text{Rally Duration (s)}}{\text{Shots per Rally}}$$

If events arrive too quickly, the actuation might act as consecutive vibration, and the human tactile system cannot fully distinguish them as separate impulses. The intervals between stimuli plays a critical role in whether vibrotactile patterns are perceived correctly (Yeganeh et al., 2025). The ISI therefore sets a practical window for haptic actuator design.

3.2 Cross-Sport Temporal Comparison

Table 1 presents a comparison of temporal metrics across four elite-level racket sports, compiled from published sports science literature.

Table 1: *Cross-sport temporal metrics at elite singles level*

Sport/Surface	**RallyDuration(s)**	**ShotsPerRally**	**DerivedISI(s)**	**ShotFrequency(s^{-1})**
Table Tennis	2.5 – 4.0	3.0 – 7.0	0.57 – 0.83	1.20 – 1.75
Badminton	5.5 – 12.9	5.4 – 12.3	1.02 – 1.05	0.95 – 0.98
Tennis(Hard Court)	4.9 – 6.5	3.4 – 5.3	1.22 – 1.44	0.69 – 0.82
Tennis(Clay Court)	6.2 – 8.1	3.4 – 6.7	1.21 – 1.82	0.55 – 0.83

Sources: Table tennis — Ebatt Table Tennis Analytics (2023); Badminton — Laffaye et al. (2015); Tennis — Pluim et al. (2023). ISI values are derived as Rally Duration / Shots per Rally.

3.2.1 *Table Tennis*

Ebatt Table Tennis Analytics (2023) report that a typical professional rally lasts 3–7 strokes, with each stroke played within 0.57–0.83 seconds, 1.20 – 1.75 strokes per second. This gives the shortest ISI of all four sports, making the actuator rate too fast for comfortable tactile discrimination.

3.2.2 *Badminton*

Laffaye et al. (2015) tracked elite Olympic singles finals and found shot frequency increased by 34% over two decades, reaching 1.3 s^{-1} by recent years. Despite longer rallies than tennis, the high stroke rate produces an ISI below 1 second, still too fast for reliable haptic resolution.

3.2.3 Tennis

Pluim et al. (2023) found in the international-level man's game, the surface type significantly influences rally duration, with hard court rallies averaging 4.9–6.5 seconds and clay court rallies extending to 6.2–8.1 seconds. The slower pace on clay results from the surface's high friction dampening ball speed, pushing players further behind the baseline and extending each inter-shot interval to approximately 1.21-1.82 s — the most favourable range for haptic actuator design among the four sports examined.

3.3 Haptic Engineering Implications and Sport Choice

At an ISI of approximately 0.7 s (table tennis) or 1.03 s (badminton), the actuator risks operating near the sensory integration limit of tactile mechanoreceptors, causing events to blur together. Clay-court tennis, with an ISI of approximately 1.21–1.82 s, provides sufficient separation to ensure each impulse is perceived and resolved before the next arrives.

A longer ISI also benefits wireless delivery. The $\sim$1.52 s window on clay court provides a substantial buffer for jitter compensation when haptic trigger packets are transmitted via a network protocol to the Android cell phone.

For these reasons, **clay-court tennis** is selected as the primary target sport for this project. It offers the most favourable temporal conditions for a first implementation of the haptic feedback system.

4 Approach

4.1 System Overview

The system processes the audio track of a tennis video to detect events, and drives smartphone vibration in response. The input video is sourced from ETV broadcast recordings; the audio track is extracted and processed offline before playback.

The overall pipeline is:

Tennis video → extract audio → filter + peak detection → feature extraction → event classification → Android vibration → spectator feels each shot

4.2 Detection Pipeline

The detection pipeline is kept general at this stage, because the specific method for identifying strikes has not yet been finalised. The planned stages are:

1. **Event annotation:** Find time points where impact-like changes occur in the sound.

2. **Feature extraction:** For each candidate time point, compute simple audio features. Feature extraction will draw on approaches established in the literature (Baughman et al., 2019; Caprioli et al., 2025), such as MFCCs, zero-crossing rate, and spectral statistics. The final feature set will be determined experimentally during implementation.

3. **Event decision:** Apply one or more classification or rule-based approaches to decide whether each candidate point is a strike, a bounce, or noise. The specific method will be chosen based on early experiments and ease of implementation.

4.3 Proof-of-Concept Extension(Optional)

If the main aim is completed well ahead of schedule, a proof-of-concept extension will be attempted: deriving rally pace from inter-event timing (Caprioli et al., 2024) and using it to drive a slow thermal signal via the Embr Wave 2. No user study is planned for this extension. It is included here for completeness; whether it is feasible within the project timeline will become clear during implementation.

4.4 Actuator Design

Consumer devices are chosen over custom haptic hardware to keep the system accessible to any spectator without additional equipment.

Android smartphone (vibration — primary): An customised video player program will play the tennis video for the spectator and send haptic events to the paired Android smartphone. The smartphone is chosen as vibration hardware because it requires no extra equipment and is already in the spectator's hand (Gómez-Monroy et al., 2026).

Embr Wave 2 (proof-of-concept extension only): If the main aim is completed early, the Embr Wave 2 would be used as a thermal output for the optional extension described in Section 4.3. It is not part of the core system or the user study.

4.5 Evaluation Plan

Evaluation will cover both the detection side and the user experience:

- **Detection accuracy:** Detected events will be compared against ground-truth labels from video analysis. Metrics: accuracy, precision, recall, and F1-score (Baughman et al., 2019).
- **Haptic experience:** A small user study (approximately 10–20 participants) will be conducted once the detection system is implemented. Each participant will watch short clips of clay-court tennis under two conditions: *video + audio only* and *video + audio + vibration feedback*. The order of conditions will be counterbalanced across participants. After

each condition, participants will complete a short questionnaire rating their sense of immersion, presence, and enjoyment on 7-point Likert scales, following similar user study designs used in the literature (Iekura et al., 2015; Peng, 2022). Basic demographic information (e.g., age, prior tennis-watching experience) will also be collected. Paired-sample statistical tests will be used to compare ratings between conditions.

4.6 Audio Quality and Noise Considerations

The audio signal in a real-world deployment would ideally come from dedicated court-level microphones, providing a clean feed of ball impact and bounce sounds. However, the video recordings sourced from ETV for this project are broadcast recordings, which contain additional sound sources: crowd noise, commentator voices, and ambient stadium sounds. These can overlap with or mask the acoustic signatures of ball events (Baughman et al., 2019; Caprioli et al., 2025).

This is a practical limitation of the study. In tennis, crowd noise is generally lower in intensity and more intermittent than in sports such as football or indoor basketball (Caprioli et al., 2025). The detection system will be tested on the ETV broadcast recordings to assess whether this noise causes significant interference. If the system performs reliably despite broadcast noise, this itself constitutes a useful practical finding. If noise proves problematic, filtering strategies or a more noise-robust feature set will be explored.

5 Progress

5.1 Literature Review

A broad literature review has been completed across two areas. First, haptic feedback systems for sports spectators and athletes were surveyed, covering dedicated vibration devices (de Graaf, 2022; Iekura et al., 2015; Sabnis et al., 2025), custom haptic interfaces (Mizushina et al., 2015; Takahashi et al., 2021), thermal actuators (Ooms et al., 2023; Peng, 2022), smartphone-based feedback (Gómez-Monroy et al., 2026), and a comprehensive review of wearable haptic actuator technologies (Yin et al., 2021). Second, acoustic detection methods for tennis were reviewed in depth (Baughman et al., 2019; Caprioli et al., 2024, 2025). The review confirmed the gap that motivates this project: no existing work combines audio-based detection with accessible haptic feedback for remote sports spectators.

5.2 Development Environment

Python libraries for project have been identified: `librosa` for audio feature extraction, `PySide6` for the user interface of video player. For the Android phone actuator, the `Vibrator` API will be used. A research API for the Embr Wave 2 has been obtained for the optional proof-of-

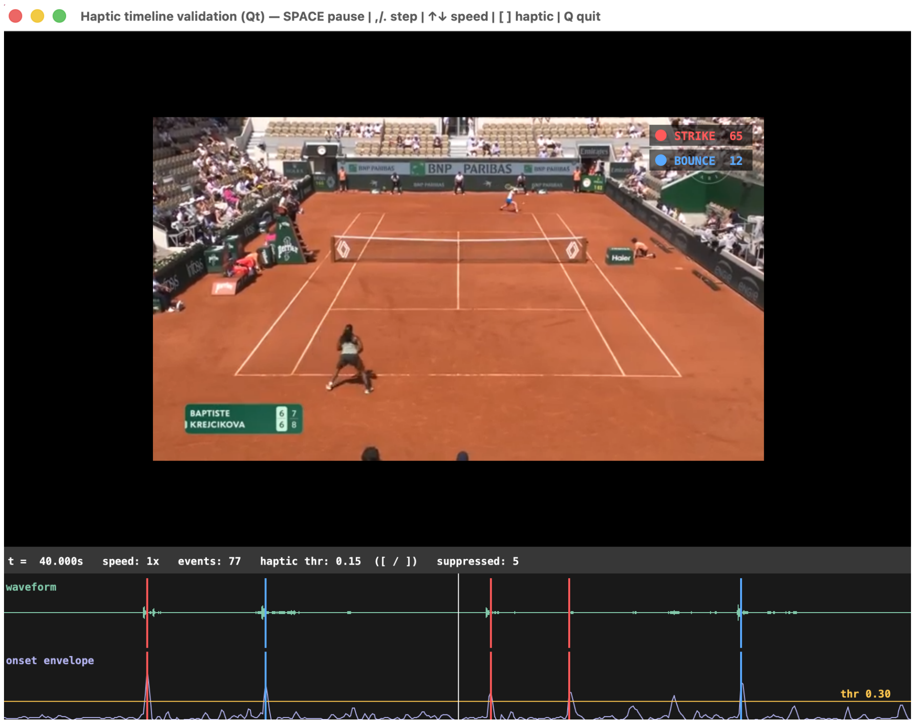

Figure 1: Haptic Video Player Prototype

concept extension; initial investigation of its limitations (long duration for actuation) confirms it is not suitable as a core deliverable.

A prototype video player has been developed. The player loads a tennis match video and displays a time-axis alongside the playback, on which detected events are marked. The input video corpus for this project will be sourced from ETV (Educational Television, New Zealand), which provides access to broadcast recordings of sports events including clay-court tennis matches. Two test videos have been collected from ETV for initial evaluation. These are broadcast recordings with stereo audio containing the typical mix of court sounds, crowd noise, and commentary. If the available ETV recordings are insufficient in quantity or variety, additional clay-court tennis match recordings will be sourced from publicly available online archives (e.g., YouTube broadcast recordings), or requested directly from ETV through the university access channel. No event detection method has been finalised yet. Figure 1 shows a screenshot of the prototype.

5.3 Changes from the Original Proposal

The scope has been refined in three ways since the original proposal. First, the sensor approach has been changed from general hardware to audio-based detection. Second, the actuator design has been focused on the Android smartphone as the sole core actuator; the Embr Wave 2 is retained only as a potential proof-of-concept extension. Third, the sport selection has been justified through the cross-sport ISI analysis in Section 3, with clay-court tennis identified as the optimal option.

6 Plan / Schedule

Table 2 shows the planned work for Semester B.

Table 2: *Project plan*

Period	**Task**
Week 1–2 (13–26 Jun)	Collect audio recordings from tennis matches; annotate events; prepare draft STEM ethics application
Week 3–4 (27 Jun–10 Jul)	Investigate and select the detection approach for events; **Submit STEM ethics application (deadline 6 Jul 2026)**
Week 5–7 (11–31 Jul)	Implement and refine the detection system; integrate with Android vibration app; run pilot tests and finalise user study materials (clips, questionnaire, instructions)
Week 8–9 (1–14 Aug)	**Conduct user study as soon as ethics approval is received**; collect questionnaire data
Week 10–11 (15–28 Aug)	Analyse user study results; make any final adjustments to the system
Week 12 (29 Aug–4 Sep)	Finalise system; prepare project presentation
4 Sep 2026	**Project presentation**
Week 13–16 (5 Sep–2 Oct)	Write and refine dissertation
Week 17–19 (3–23 Oct)	Complete dissertation
23 Oct 2026	**Dissertation submission**

Risk 1 — Data collection: Data collection requires sufficient clay-court tennis match recordings with usable broadcast audio. The primary source is ETV; if recordings are insufficient, additional material will be sourced from publicly available online archives or requested directly from ETV through the university access channel.

Risk 2 — Ethics approval timing: The user study cannot begin until ethics approval is granted. The University of Waikato STEM Ethics Committee meets monthly; the 6 July 2026 deadline is the critical one — approval is expected within 10 working days (approximately 22 July), leaving sufficient time before the Week 10–11 user study window. If the application misses the 6 July deadline, the next round closes 3 August and approval would arrive too late for the planned study dates.

Risk 3 — Proof-of-concept extension: The optional thermal extension depends on completing the main aim early, on the Embr Wave 2 API providing sufficient control, and on having time remaining after the user study. All three conditions are uncertain; the extension may not be attempted at all.

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A Project Proposal

Home Sports Viewing with Wearable Heat and Vibration Feedback

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Abstract

Previous *Feel the Sport* work showed that wrist-worn vibration derived from athlete motion data can make watching sport more engaging, but feedback was limited to a single modality (vibration) and synchronized playback rather than real-time broadcasting (Nichols et al., 2026; Poutoa, 2025). This project proposes an affordable, home-oriented wearable that combines vibrotactile and thermal feedback to enhance sports viewing. Using a racket sport for the test, the system will map impact events to vibrations and game intensity to warming/cooling patterns. A user study will be taken, focusing on immersion, enjoyment, comfort, and synchronization. As a technical sub-goal, Wi-Fi-based (e.g., MQTT) communication will be prototyped to explore the feasibility of future real-time haptic broadcasting. AI-based detection of impact sounds from audio will be noted as future work rather than implemented in this project.

Introduction

Watching sport at home is mostly an audio-visual experience. Earlier *Feel the Sport* studies used a LilyGo T-Watch to capture motion from tennis and map it to vibrations during synchronized playback; participants reported that haptics increased immersion and were rarely distracting (Nichols et al., 2026; Poutoa, 2025). A later study extended this to multiple sports using manually authored haptic tracks, again finding generally positive reactions from viewers (Nichols et al., 2026).

These results suggest that haptic is beneficial for enhancing sports spectatorship, but there are still gaps. First, the feedback so far is vibration-only, delivered on one wrist. Second, the current system operates as record–process–playback; BLE throughput and timing issues prevented a real-time implementation (Poutoa, 2025). Third, synchronization has been identified as a significant influence on user experience: small latency between video and haptics can reduce experience effect. Recent reviews of wearable haptics show that multi-sensory devices could support richer experiences while remaining lightweight and body-centred (Fleck et al., 2025; Turchet et al., 2020).

Thermal feedback is attractive for a home device because it can convey the “heat” or intensity of play without high forces. Racket sports are a good option: clear impact events support vibrotactile cues, while changes in game intensity can drive slower changes in temperature. Tennis provides direct continuity with previous work; badminton is an alternative where shuttlecock impacts occur only at the racket, simplifying event detection.

Objectives

This project aims to design and evaluate a multimodal haptic wearable for home sports spectators. The main objectives are:

- To develop a prototype device that combines a vibration motor and a Peltier-based thermal element, controlled by a low-cost microcontroller suitable for home use.
- To integrate this device with a pipeline that captures motion data from a racket sport (tennis or badminton), detects impact and intensity features, and synchronizes haptic feedback with sports video.
- To conduct a user study, comparing three viewing conditions (no haptics, vibration only, vibration+thermal) and measuring immersion, enjoyment, comfort, and perceived synchronization.
- To prototype a simple event stream and measure end-to-end latency as an exploration of future real-time haptic broadcasting(e.g., MQTT + WiFi).

Method

- **Ethics:** Before any user-facing activity, a Human Research Ethics Application will be prepared and submitted. The application will address thermal safety (maximum skin-contact temperatures and exposure times), electrical safety, informed consent, recruitment, data handling, and participants' right to withdraw without penalty. No user study will begin until formal approval is granted.
- **Sport and data capture:** A player will wear an accelerometer-equipped device during short rallies in the chosen racket sport. Sensor data and video will be recorded simultaneously. A clapper-style gesture will mark a shared start point for alignment, following earlier *Feel the Sport* methodology.
- **Hardware design:** A small, wrist or forearm-mounted module will contain a Peltier element and a vibration motor. An microcontroller will drive both actuators and communicate with a laptop via BLE or Wi-Fi.
- **Mapping and synchronization:** Peaks in acceleration magnitude will be used to detect impacts and trigger short vibration pulses. Game intensity or key moments will be estimated to be mapped to gradual warming or cooling. Because temperature changes slowly, thermal cues will be used as background state. Haptic events will be scheduled against

the video timeline during playback.

- **User study:** After testing and ethics approval, a within-subjects study with approximately 15–20 adult participants will be run in a lab arranged like a living room. Each participant will watch the same clip(s) under three conditions: no haptics, vibration only, and vibration+thermal. Questionnaires will capture perceptions of fit with other senses, liking of haptics, distraction, variation, thermal comfort, and synchronization.
- **Real-time sub-goal:** A prototype using Wi-Fi and MQTT will send "hit" events to the wearable in real time in a single-room setup. End-to-end latency will be measured and compared qualitatively with earlier BLE. This is exploratory and not part of the main user study.
- **Future work:** Future extensions could explore AI models for sound detection from audio to trigger haptics.

Schedule

- Weeks 1–4: Finalize sport choice (tennis or badminton) with supervisors. Prepare and submit Division of STEM Full Human Research Ethics Application for the user study and device testing.
- Weeks 5–9: Await ethics decision; acquire hardware and implement basic vibration and safe thermal control.
- Weeks 10–11: After ethics approval, capture motion and video data with volunteers; implement event detection and initial mapping to vibration and thermal patterns; integrate synchronized playback.
- Weeks 12–15: Testing of mappings and synchronization. Finalize user study materials (questionnaires, interview prompts) under ethics approval.
- Weeks 16–20: Run the main user study (no haptics vs vibration vs vibration+thermal) in home-like lab conditions; collect questionnaire data.
- Weeks 21–23: Analyze quantitative and qualitative results; interpret findings in relation to previous *Feel the Sport* work and wearable haptics literature.
- Weeks 24–25: Implement and test the exploratory Wi-Fi/MQTT real-time prototype and latency measurements; prepare slides and materials for the project presentation.
- Weeks 26–30: Polish written material into the final dissertation, incorporating supervisor feedback and deeper discussion of limitations and future work (including possible AI/audio-based impact detection).

Resources

- Hardware: ESP32-based wearable (e.g., LilyGo T-Watch), Peltier modules and drivers, vibration motors, batteries and enclosures, and tennis or badminton equipment plus an indoor space for data capture.
- Software: Arduino IDE and ESP32 toolchain for firmware; Python for data processing and playback; MQTT broker and client library for the real-time prototype; Overleaf/LaTeX for documentation.
- Human resources: Supervisor guidance on design, ethics, and analysis, and participants for pilot and main user studies.

Appendix References

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